

Workforce & Employee Analytics Intelligence System

Project Overview

Organizations today generate a large amount of employee-related data, but many HR teams struggle to convert that data into meaningful workforce insights. Challenges such as employee attrition, overtime stress, low work-life balance, and employee dissatisfaction can directly affect productivity and long-term business growth.

This project focuses on analyzing workforce and employee data to better understand the factors influencing employee retention, engagement, and overall workforce performance. The project uses the IBM HR Analytics Employee Attrition dataset to identify patterns related to employee turnover, department trends, job roles, work-life balance, overtime, and satisfaction levels.

The main objective of this project is to help HR teams and management make informed decisions through analytics, reporting, and visualization. By analyzing workforce data, the system aims to provide insights that can support employee retention strategies, improve workforce planning, and enhance employee experience within the organization.

The project combines business analytics and HR analytics concepts using tools such as Excel, SQL, Python, Power BI, Jira, and Confluence. Jira is used to manage project workflow and sprint planning, while Confluence is used for project documentation and reporting.

Key Objectives

- Analyze employee attrition trends and workforce patterns
- Identify factors affecting employee satisfaction and retention
- Build dashboards and reports for workforce insights
- Support HR decision-making using analytics
- Improve understanding of workforce behavior through data analysis

Dataset Information

The project uses a cleaned HR employee dataset containing information such as:

- Employee age
- Department
- Job role
- Monthly income
- Work-life balance
- Overtime

- Job satisfaction
- Performance ratings
- Years at company
- Attrition status

The dataset helps simulate real-world HR analytics scenarios commonly faced by organizations.

Project Workflow Tracking

Type	Key	Summary	Assignee	Priority	Status	Updated
☒	WOR-21	Prepare final presentation		Medium	DONE	Jun 24, 2026, 14:44
☒	WOR-20	Final project reporting		Medium	DONE	Jun 24, 2026, 14:44
☒	WOR-19	Generate workforce recommendations		Medium	DONE	Jun 24, 2026, 14:01
☒	WOR-18	Build attrition prediction model		Medium	DONE	Jun 21, 2026, 09:48
☒	WOR-17	Generate engagement insights		Medium	TO DO	May 22, 2026, 07:52
☒	WOR-16	Work-life balance analysis		Medium	TO DO	May 22, 2026, 07:42
☒	WOR-15	Track performance ratings		Medium	TO DO	May 22, 2026, 07:42
☒	WOR-14	Analyze employee satisfaction		Medium	DONE	Jun 21, 2026, 09:47
☒	WOR-13	Generate hiring metrics		Medium	DONE	Jun 24, 2026, 14:01
☒	WOR-12	Workforce distribution analysis		Medium	DONE	Jun 21, 2026, 09:47
☒	WOR-11	Create recruitment dashboard		Medium	DONE	Jun 24, 2026, 14:01
☒	WOR-10	Analyze hiring trends		Medium	DONE	Jun 24, 2026, 14:01
☒	WOR-9	Build retention insights		Medium	DONE	Jun 24, 2026, 14:01
☒	WOR-8	Create attrition KPIs		Medium	DONE	Jun 24, 2026, 14:01
☒	WOR-7	Clean employee data		Medium	DONE	May 23, 2026, 07:07
☒	WOR-6	Analyze attrition trends		Medium	DONE	May 30, 2026, 07:05
☒	WOR-5	Collect HR dataset		Medium	DONE	May 22, 2026, 07:57
	WOR-4	Predictive Workforce Analytics		Medium	TO DO	May 22, 2026, 07:30
	WOR-3	Employee Performance Insights		Medium	TO DO	May 22, 2026, 07:29
	WOR-2	Recruitment Analytics Dashboard		Medium	TO DO	May 22, 2026, 07:28
	WOR-1	Employee Attrition Analysis		Medium	TO DO	Jun 24, 2026, 14:03

